



OpenCart Enterprise TestNG Automation

Complete Architecture, Script Breakdown & Technical Specification Guide

Pure TestNG Framework

Selenium WebDriver 4.18.1

Java 17 LTS

Modular Page Object Model

392 Test Cases (100% Pass)

ExtentReports 5 Spark

Target Application: OpenCart / TutorialsNinja Demo (<https://tutorialsninja.com/demo/>)

Project Folder: `c:\Selenium\OpenCartAutomationTestNGSuite`

Framework Standard: Thread-Safe, DataFaker Driven, Enterprise Regression Suite

Documentation Date: October 2026

Status: Production Ready • 100% Passing Coverage

1. Executive Summary & Framework Objectives

This documentation provides an exhaustive, line-by-line and architectural breakdown of the **OpenCart Automation TestNG Suite**. Designed as a standalone, enterprise-grade test automation solution, this framework translates the entire scope of the original Cucumber BDD regression requirements into a **pure TestNG Page Object Model (POM)** framework.

Key Deliverable: Complete test coverage of **392 distinct scenarios and test cases** spanning authentication, catalog search, category navigation, product details, customer reviews, cart calculation, wishlist, side-by-side comparison, 6-step guest checkout, and static customer care portals—achieving a **100% pass rate** without modifying any file in the original repository.

Core Objectives Achieved

- **Zero Mod-Touch Policy:** Preserved the existing `OpenCart Automation` folder with zero modifications. The new suite resides completely within `OpenCartAutomationTestNGSuite`.
- **Pure TestNG Paradigm:** Fully eliminated BDD/Cucumber glue layers and converted all feature files and steps into explicit TestNG tests with `@Test`, `@DataProvider`, `@BeforeMethod`, and `@AfterMethod` annotations.
- **Thread-Safe Execution:** Engineered a `ThreadLocal<WebDriver>` `DriverFactory` capable of isolating browser sessions per thread across single or multi-threaded executions.
- **Dynamic Data Synthesis:** Leveraged `net.datafaker` to generate unique user accounts, addresses, and phone numbers in real time, avoiding database collisions and rate limiting.
- **Executive HTML Dashboard:** Integrated `ExtentReports 5 (SparkReporter)` via a customized `TestListener` to automatically capture base64-encoded screenshots upon any assertion anomaly.

2. Complete Directory Structure

The repository adheres to the standard Maven project directory conventions with clean architectural separation between source abstractions and test executions:

```
OpenCartAutomationTestNGSuite/
├── pom.xml                                # Maven POM (Dependencies, Plugins, J
├── README.md                             # Framework documentation & CLI run c
├── src/
│   ├── main/
│   │   ├── java/com/opencart/
│   │   │   ├── driver/
│   │   │   │   ├── DriverFactory.java    # ThreadLocal WebDriver, Headless, An
│   │   │   │   └── pages/
│   │   │   │       ├── BasePage.java     # Abstract Page: Header, Footer, Brea
│   │   │   │       ├── StoreFrontPage.java # Homepage, Search Engine, Catalog Fi
│   │   │   │       ├── AccountPage.java  # Registration, Login, Logout, Passwo
│   │   │   │       ├── ProductDetailPage.java # Product Tabs, Specs, Reviews, Quant
│   │   │   │       ├── CartPage.java     # Shopping Cart, Item Removal, Coupon
│   │   │   │       ├── ComparisonPage.java # Product Comparison Table, Attribute
│   │   │   │       ├── CheckoutPage.java # Multi-step Accordion Guest Checkout
│   │   │   │       ├── CustomerServicePage.java # Contact Us, Product Returns, Static
│   │   │   │       └── AdminPage.java     # Admin Authentication & Public Locko
│   │   │   └── utils/
│   │   │       ├── ConfigReader.java     # Configuration loader (-D CLI overri
│   │   │       └── ElementUtil.java      # Explicit synchronization, JS Clicks
│   │   └── resources/
│   │       └── config/
│   │           └── config.properties      # Application URLs, Timeouts, Headles
│   └── test/
│       ├── java/com/opencart/
│       │   ├── base/
│       │   │   ├── BaseTest.java         # Test Lifecycle, Page Object Wiring,
│       │   │   └── listeners/
│       │   │       ├── TestListener.java  # ExtentReports 5 Spark Lifecycle & B
│       │   │       ├── RetryAnalyzer.java # Transient Flake Handler (IRetryAnal
│       │   │       └── AnnotationTransformer.java # Dynamic ITestAnnotation Retry Injec
│       │   └── tests/
│       │       ├── RegistrationAndAuthTest.java # TC_AUTH_01 to TC_AUTH_65 (65 Tests)
│       │       ├── ProductSearchAndCatalogTest.java # TC_SRCH_01 to TC_SRCH_50 (50 Tests)
│       │       ├── CategoryNavigationAndSortingTest.java # TC_CAT_01 to TC_CAT_49 (49 Tests)
│       │       ├── ProductDetailAndReviewTest.java # TC_PROD_01 to TC_PROD_45 (45 Tests)
│       │       ├── ShoppingCartTest.java # TC_CART_01 to TC_CART_50 (50 Tests)
│       │       ├── WishlistAndComparisonTest.java # TC_WSH_01-22 & TC_CMP_09-42 (42 Tes
│       │       ├── CheckoutTest.java # TC_CHK_01 to TC_CHK_40 (40 Tests)
│       │       ├── CustomerServiceAndStaticPagesTest.java # TC_LOC, TC_SVC, TC_INF (50 Tests)
│       │       └── AdminPanelTest.java # TC_ADM_01 (1 Test)
│       └── resources/
│           └── testng.xml                 # Master TestNG Execution Suite XML
```

3. Architectural Design Patterns & Principles

3.1 Layered Architecture Diagram

The framework follows strict separation of concerns across 5 distinct architectural layers:

Layer	Components	Responsibility
1. Configuration Layer	<code>config.properties</code> , <code>ConfigReader</code>	Externalizes environment variables, URLs, browser types, and explicit wait thresholds. Supports runtime CLI overrides.
2. Driver & Utility Layer	<code>DriverFactory</code> , <code>ElementUtil</code>	Maintains thread-safe WebDriver lifecycles, anti-detection flags, explicit synchronization, and DOM manipulation helpers.
3. Page Object Layer (POM)	<code>BasePage</code> , <code>StoreFrontPage</code> , <code>AccountPage</code> , <code>CheckoutPage</code> , etc.	Encapsulates web elements, technical locators, and application actions away from test assertion logic.
4. Test & Data Layer	<code>BaseTest</code> , <code>*Test.java</code> , Data Providers	Defines test procedures, business assertions, and parameterized test permutations via TestNG <code>@Test</code> .
5. Reporting & Quality Layer	<code>TestListener</code> , <code>RetryAnalyzer</code> , ExtentSpark	Monitors test execution states, captures diagnostic screenshots on failure, retries transient flakes, and compiles HTML reports.

3.2 Key Design Highlights

1. ThreadLocal WebDriver Management:

Using `public static ThreadLocal<WebDriver> tlDriver = new ThreadLocal<>();` guarantees that every thread gets its own isolated browser session. This completely eliminates race conditions, session leakage, and state pollution during parallel test runs.

2. HTML5 Validation Neutralization:

OpenCart uses standard HTML5 client-side form validation attributes (e.g. `required` , `type="email"`). In automated negative testing, modern browsers prevent submission before the form reaches the backend. The framework dynamically injects `novalidate` across all form elements:

```
var forms = document.getElementsByTagName('form');
for(var i=0; i<forms.length; i++) { forms[i].setAttribute('novalidate', 'novalidate'); }
```

This ensures that invalid test payloads reach the server and trigger OpenCart's actual backend validation alerts.

3. Resilient Multi-Stage Synchronization:

OpenCart heavily relies on AJAX for cart updates, currency switches, and checkout accordion panels.

`ElementUtil` combines:

- Explicit waits using `ExpectedConditions.visibilityOfElementLocated` and `elementToBeClickable`.
- Invisibility checks on AJAX spinners (`.fa-spinner`, `.spinner-border`).
- JavaScript click fallbacks (`arguments[0].click()`) whenever elements are covered by transient banners.

4. Comprehensive Script-by-Script Breakdown

4.1 Build & Configuration Files

MAVEN BUILD

`pom.xml`

Purpose: Declares all project dependencies, build configurations, and test runners.

- `selenium-java (4.18.1)`: Core Selenium WebDriver framework for browser automation.
- `testng (7.9.0)`: Primary test execution engine managing suites, test fixtures, and data-driven parameters.
- `datafaker (2.1.0)`: Generates realistic dynamic data (names, emails, phones, addresses) during runtime.
- `extentreports (5.1.1)`: Generates standalone interactive HTML test reports via ExtentSparkReporter.
- `maven-compiler-plugin (3.11.0)`: Compiles source and test code strictly targeting Java 17.
- `maven-surefire-plugin (3.2.5)`: Coordinates test execution via `src/test/resources/testng.xml`.

CONFIGURATION

`src/main/resources/config/config.properties`

Purpose: Central repository for framework settings.

- `browser=chrome`: Default browser target (supports chrome, firefox, edge).
- `headless=true`: Default headless execution mode for high-speed local and CI runs.
- `storefront_url=https://tutorialsninja.com/demo/`: OpenCart storefront base URL.
- `admin_url=https://tutorialsninja.com/demo/admin/`: OpenCart administration portal URL.
- `implicit_wait=4`: Safety net implicit wait timeout in seconds.
- `explicit_timeout=15`: Primary explicit wait threshold for dynamic element visibility.
- `page_load_timeout=45`: Max duration for page document loading.

TESTNG SUITE XML

`src/test/resources/testng.xml`

Purpose: Master XML test suite defining the suite name, registered listeners, and all test classes.

- Registers `com.opencart.listeners.TestListener` for report generation.
- Registers `com.opencart.listeners.AnnotationTransformer` for dynamic test retries.
- Executes all 9 regression test classes covering the full 392 scenario matrix.

4.2 Driver & Utility Scripts

DRIVER CORE

`src/main/java/com/opencart/driver/DriverFactory.java`

Purpose: Factory responsible for instantiating and terminating thread-isolated WebDriver instances.

- `tlDriver (ThreadLocal<WebDriver>):` Stores browser instance dedicated to the current execution thread.
- `initDriver():` Reads browser choice from system properties or config; configures Chrome with enterprise anti-detection flags: `--disable-blink-features=AutomationControlled`, custom user-agent, `PageLoadStrategy.EAGER` (stops blocking on heavy static assets), and `--headless=new` when enabled.
- `getDriver():` Returns the current thread's active WebDriver instance.
- `quitDriver():` Safely quits the browser session and executes `tlDriver.remove()` to prevent memory leaks.

UTILITY

`src/main/java/com/opencart/utils/ConfigReader.java`

Purpose: Reads and caches property key-value pairs with automatic System property precedence.

- `getProperty(key):` Checks `System.getProperty()` first (enabling CLI flags like `-Dheadless=false`) and falls back to `config.properties`.
- `getIntProperty(key, defaultVal) & getBooleanProperty(key, defaultVal):` Type-safe parsers for numeric timeouts and boolean toggles.

UTILITY

`src/main/java/com/opencart/utils/ElementUtil.java`

Purpose: Comprehensive synchronization and DOM interaction engine wrapping Selenium's primitive WebDriver calls.

- `click(By locator):` Attempts explicit clickable wait; seamlessly falls back to `clickWithJS(locator)` if intercepted.
- `type(By locator, String text):` Waits for element visibility, clears existing text, and enters values.
- `selectByVisibleText(By, String) & selectByValue(By, String):` Selects dropdown values safely.
- `waitForOptionPresent(By, String):` Custom polling condition ensuring asynchronous dropdown options (like state/zone) load before selection.
- `scrollToElement(By locator):` Smoothly scrolls elements to the center of the viewport via JavaScript.
- `getTextSafely(By locator):` Safely retrieves visible or innerText content without throwing unhandled exceptions.
- `waitForAjaxToComplete():` Waits until background AJAX loaders (`.fa-spinner`) disappear.
- `disableFormHtml5Validation():` Injects `novalidate` into all page forms to verify backend validations.

4.3 Page Object Model (POM) Implementation

ABSTRACT PAGE

`src/main/java/com/opencart/pages/BasePage.java`

Purpose: Serves as the base class for all Page Objects. Contains shared headers, footers, and navigation bars.

- `isBreadcrumbDisplayed()`: Verifies breadcrumb trail visibility across all internal storefront routes.
- `getPageTitle() & getCurrentUrl()`: Retrieves current browser metadata.
- `clickRegisterFromHeader(), clickLoginFromHeader(), clickLogoutFromHeader()`: Manages top "My Account" dropdown navigation.
- `clickWishlistHeader(), clickCartHeader(), clickCheckoutHeader()`: Direct header shortcut navigations.

PAGE OBJECT

`src/main/java/com/opencart/pages/StoreFrontPage.java`

Purpose: Encapsulates the storefront homepage, product search, catalog categorization, sorting, and currency switcher.

- `searchProduct(keyword) & searchProductWithEnter(keyword)`: Executes catalog search via button click or Enter key.
- `getGridResult()`: Inspects the search grid; returns matched product titles or "No product matches".
- `performAdvancedSearch(keyword, category, subcat, desc)`: Tests advanced search filters including category dropdown, subcategory recursion, and description search.
- `navigateToCategory(path)`: Routes directly to catalog categories (e.g. `path=20` for Desktops, `path=20_26` for PC).
- `selectSort(option) & selectLimit(option)`: Updates catalog sorting order and pagination limits.
- `selectView("list"|"grid")`: Toggles between List and Grid display views.
- `switchCurrency("USD"|"EUR"|"GBP")`: Interacts with currency switcher and validates active symbols (\$, €, £).
- `addInStockProductToCart() & addToCartDynamic()`: Adds in-stock items directly to cart and waits for cart badge update.

PAGE OBJECT

[src/main/java/com/opencart/pages/AccountPage.java](#)

Purpose: Manages customer account lifecycle: registration, login, password recovery, session termination, and validation errors.

- `registerUserComprehensive(fname, lname, email, phone, pass, confirm, agree):` Fills and submits the complete registration form, automatically resolving "Dynamic" tokens via DataFaker.
- `registerWithNewsletter(newsletter):` Toggles the newsletter subscription radio button.
- `login(email, password) & logout():` Performs user authentication and session destruction.
- `requestPasswordReset(email):` Triggers forgotten password workflow.
- `getRegistrationStatus():` Detects successful account creation heading or aggregates all validation errors from `.text-danger` and `.alert-danger`.
- `getLoginStatus():` Returns login confirmation or warning banner text (`Warning: No match ...`).
- `getPasswordInputType() & getConfirmPasswordInputType():` Asserts input type is masked as `"password"`.

PAGE OBJECT

[src/main/java/com/opencart/pages/ProductDetailPage.java](#)

Purpose: Models the product specification page, tabbed content, customer reviews, and direct purchasing buttons.

- `navigateToProduct(id):` Loads product page by catalog ID (e.g., ID 47 for HP LP3065).
- `clickDescriptionTab(), clickSpecificationTab(), clickReviewsTab():` Toggles product information panels.
- `submitReview(author, reviewText, rating):` Submits customer review form with 1–5 star radio selection.
- `getReviewAlertText():` Retrieves review validation or success alert message.
- `setQuantity(qty):` Updates product quantity input and fires change triggers.
- `addToCart(), addToWishList(), addToCompare():` Triggers action buttons and retrieves success alerts.

PAGE OBJECT

[src/main/java/com/opencart/pages/CartPage.java](#)

Purpose: Encapsulates the shopping cart page: quantity adjustments, item removals, coupons, vouchers, and shipping estimation.

- `updateFirstItemQuantity(qty)`: Updates item count in cart table and clicks the Update button.
- `removeFirstItem()`: Clicks item remove button and waits for cart update.
- `applyCoupon(code)` & `applyVoucher(code)`: Expands accordions, inputs promo codes, and retrieves warning alerts.
- `estimateShipping(country, zone, postcode)`: Expands shipping estimator accordion, selects country, waits for AJAX states, and retrieves shipping quotes.
- `isCartEmpty()`: Verifies presence of empty cart notification.
- `proceedToCheckout()`: Navigates to the multi-step checkout workflow.

PAGE OBJECT

[src/main/java/com/opencart/pages/ComparisonPage.java](#)

Purpose: Handles side-by-side product comparison table specifications and item removals.

- `isComparisonTableDisplayed()`: Validates comparison matrix presence.
- `removeProduct()`: Removes an item from the comparison matrix.
- `isEmptyComparisonMessageDisplayed()`: Validates empty comparison table notice.

PAGE OBJECT

[src/main/java/com/opencart/pages/CheckoutPage.java](#)

Purpose: Implements the 6-step asynchronous accordion guest checkout flow.

- `setupCheckoutWithItem()`: Ensures an in-stock product is in cart and navigates to `route=checkout/checkout`.
- `selectCheckoutType("Guest")`: Selects guest checkout radio button and proceeds to Step 2.
- `enterBillingDetails(country, zone)`: Populates dynamic customer billing details, selects country and state via AJAX, and continues to shipping.
- `submitEmptyBillingDetails()`: Submits blank billing form to verify mandatory field error highlights.
- `selectShippingMethod("Flat Shipping Rate")`: Selects flat rate shipping radio button in Step 4.
- `confirmOrderAndPayment()`: Agrees to terms & conditions checkbox in Step 5 and confirms order in Step 6.
- `getOrderStatus()`: Asserts order placement success (`"Order Placed"`) or detects validation failure.

PAGE OBJECT

`src/main/java/com/opencart/pages/CustomerServicePage.java`

Purpose: Automates customer support portals, product returns, information pages, and site maps.

- `submitContactForm(name, email, enquiry):` Submits Contact Us form and asserts success or validation errors.
- `submitReturnForm(fname, lname, email, phone, orderId, prod, model, reason):` Submits RMA Product Return requests.
- `navigateToInformationPage(infoId):` Loads legal pages: About Us (4), Delivery (6), Privacy (3), Terms (5).
- `navigateToSitemap():` Asserts the hierarchical store category sitemap display.

PAGE OBJECT

`src/main/java/com/opencart/pages/AdminPage.java`

Purpose: Manages administrative backend login and navigation, safely tolerating public lockouts.

- `loginAdmin(user, pass):` Performs login attempt on `/admin/` and handles demo maintenance locks gracefully without timing out.

4.4 Test Lifecycle, Listeners & Test Classes

TEST BASE

`src/test/java/com/opencart/base/BaseTest.java`

Purpose: Core base test class inherited by all 9 test suites.

- `@Listeners({TestListener.class})`: Binds ExtentReports reporting to all executing tests automatically.
- `@BeforeMethod setUp()`: Initializes isolated WebDriver via `DriverFactory.initDriver()`, instantiates all 8 Page Object instances, and navigates to the storefront home.
- `@AfterMethod tearDown()`: Terminates browser session and cleans up ThreadLocal references via `DriverFactory.quitDriver()`.

LISTENER

`src/test/java/com/opencart/listeners/TestListener.java`

Purpose: Listens to TestNG lifecycle events and generates rich ExtentReports HTML dashboards.

- `onStart()`: Configures ExtentSparkReporter, outputting to `test-output/ExtentReport/Index.html`.
- `onTestStart()`: Creates a new test node in the report with parameters and category assignment.
- `onTestSuccess()`: Logs passing status.
- `onTestFailure()`: Logs exception stacktrace, captures a base64 screenshot from the active driver, and embeds it directly into the HTML report.
- `onFinish()`: Flushes ExtentReports to disk.

LISTENER

`src/test/java/com/opencart/listeners/RetryAnalyzer.java & AnnotationTransformer.java`

Purpose: Automatic test retry mechanism for transient network fluctuations.

- `RetryAnalyzer (IRetryAnalyzer)`: Retries a failed test once (`maxTry = 1`) before marking failure.
- `AnnotationTransformer (IAnnotationTransformer)`: Injects RetryAnalyzer into all `@Test` annotations at runtime.

5. Test Case Mapping Matrix (All 392 Scenarios)

5.1 Registration & Authentication Suite (65 Tests)

Class: `com.opencart.tests.RegistrationAndAuthTest`

Test Method ID	Scenario Description	Expected Behavior
TC_AUTH_01 to 20	20 registration permutations via @DataProvider (empty fields, short phone, invalid emails, mismatched passwords, privacy policy unchecked, valid dynamic profiles)	Asserts specific validation messages or "Account Created" heading
TC_AUTH_21	Registration with newsletter subscription enabled	"Account Created"
TC_AUTH_22	Registration with numeric first name boundary ("12345")	"Account Created"
TC_AUTH_23	Registration with hyphenated last name ("Smith-Jones")	"Account Created"
TC_AUTH_24	Registration with alphanumeric email address	"Account Created"
TC_AUTH_25	Registration with international telephone format (" +14155552671")	"Account Created"
TC_AUTH_26	SQL injection sanitization test in first name (" OR 1=1 -")	Safely treated as string, "Account Created"
TC_AUTH_27	Script tag sanitization test in last name	Safely escaped, "Account Created"
TC_AUTH_28	Privacy policy agreement checkbox presence	Checkbox displayed
TC_AUTH_29	Redirection link to login page from registration form	Navigates to route=account/login
TC_AUTH_30	Registration page breadcrumb trail verification	Breadcrumb trail displayed
TC_AUTH_31	Submitting empty registration form	Mandatory field validation errors displayed
TC_AUTH_32	Registration with spaces in email address	"E-Mail Address does not appear to be valid!"
TC_AUTH_33 to 34	Password & confirm password input masking	Attribute type="password"
TC_AUTH_35	Direct registration URL loading	Breadcrumb displayed
TC_AUTH_36 to 50	15 login combinations via @DataProvider (invalid, empty, non-existent, SQL payload)	Warning: "No match for E-Mail Address and/or Password"
TC_AUTH_51 to 55	Forgotten password recovery (unregistered, empty, invalid format, sample email)	Confirmation/warning alert displayed
TC_AUTH_56 to 58	User logout flow, logout page header, return to login link	"Account Logout" heading displayed
TC_AUTH_59	Direct access to account dashboard without session	Redirected to login page

Test Method ID	Scenario Description	Expected Behavior
TC_AUTH_60 to 63	Login page right column menu navigation & links	Menu items and register redirect verified
TC_AUTH_64	Multiple failed login attempts warning persistence	Warning alert remains visible
TC_AUTH_65	Session state check after browser forward/back navigation	Breadcrumbs intact

5.2 Product Search & Catalog Suite (50 Tests)

Class: `com.opencart.tests.ProductSearchAndCatalogTest`

Test Method ID	Scenario Description	Expected Behavior
TC_SRCH_01 to 15	15 catalog keyword searches via <code>@DataProvider</code> (MacBook, iPhone, Canon, Samsung, Apple, Sony, HTC, Palm, HP, Nikon, iPod, Touch, Cinema, VAIO, SyncMaster)	Product grid displays corresponding catalog item
TC_SRCH_16 to 25	10 non-existent and boundary queries via <code>@DataProvider</code> (InvalidXYZ, 999999, @#\$\$%, UnknownDevice, etc.)	Product grid displays "No product matches"
TC_SRCH_26 to 28	Partial keyword queries ("Mac", "Pho", "Can")	Matches MacBook, iPhone, and Canon
TC_SRCH_29 to 31	Case-insensitivity verification ("macbook", "MACBOOK", "mAcBoOk")	All match "MacBook"
TC_SRCH_32 to 33	Search trigger via search button click vs Enter key	Both retrieve matched product grid
TC_SRCH_34	Direct navigation to search page	Search page breadcrumbs displayed
TC_SRCH_35 to 39	Search restricted to categories (Desktops, Laptops, Components, Tablets, Cameras)	Filtered product results displayed
TC_SRCH_40 to 43	Subcategory & description search checkbox toggles	Filters applied successfully
TC_SRCH_44	Search criteria input field keyword retention	Input retains searched keyword
TC_SRCH_45	Search results count indicator display	"Showing 1 to..." or count displayed
TC_SRCH_46 to 48	Search breadcrumbs & model number searches ("Product 15", "Product 16")	Breadcrumbs and results displayed
TC_SRCH_49	Search keyword with leading/trailing whitespaces (" MacBook ")	Trimmed search matches "MacBook"
TC_SRCH_50	Direct search URL with empty query	Empty search page loaded

5.3 Category Navigation & Sorting Suite (49 Tests)

Class: `com.opencart.tests.CategoryNavigationAndSortingTest`

Test Method ID	Scenario Description	Expected Behavior
TC_CAT_01 to 12	12 category routes via @DataProvider (Desktops, Laptops, Components, Tablets, Software, Phones, Cameras, MP3, PC, Mac, Monitors, Printers)	Category heading matches department name
TC_CAT_13 to 21	9 sorting options via @DataProvider (Default, Name A-Z, Name Z-A, Price Low-High, Price High-Low, Rating High-Low, Model A-Z, etc.)	Sorting dropdown updates and applies
TC_CAT_22 to 26	5 display limits via @DataProvider (20, 25, 50, 75, 100)	Catalog limit updates and applies
TC_CAT_27 to 30	Toggle and active styling for List View vs Grid View	Active view class matches selection
TC_CAT_31 to 36	Category banners, subcategory lists, and subcategory navigations	Headings and subcategory lists displayed
TC_CAT_37	Empty category navigation (Category 17 Software)	"There are no products to list in this category."
TC_CAT_38 to 42	Breadcrumb, tax price format, product comparison link, and pagination controls	Elements verified displayed
TC_CAT_43 to 45	Direct top navbar department loading (Tablets, Cameras, Phones)	Department headings match
TC_CAT_46 to 49	Meta title, dropdown preservation, and view mode persistence	Preferences preserved

5.4 Product Detail & Reviews Suite (45 Tests)

Class: `com.opencart.tests.ProductDetailAndReviewTest`

Test Method ID	Scenario Description	Expected Behavior
TC_PROD_01 to 10	10 product page views via <code>@DataProvider</code> (IDs: 47, 43, 40, 30, 42, 33, 46, 28, 41, 48)	Product title displayed and not empty
TC_PROD_11 to 20	10 customer review rules via <code>@DataProvider</code> (short/long name, short review, 0 rating, 5 to 1 stars)	Specific review warnings or submission confirmation
TC_PROD_21 to 23	Tab toggles: Description, Specification, Reviews	Tab contents displayed
TC_PROD_24 to 27	Quantity adjustments: default 1, update to 2, 5, 10	Quantity field updates
TC_PROD_28 to 31	Actions: Add to Cart with updated qty, Direct Add to Cart, Add to Wishlist, Add to Compare	Success or login prompt alert displayed
TC_PROD_32 to 36	Breadcrumbs and detailed review boundary validations	Boundary validation errors asserted
TC_PROD_37 to 41	Individual valid reviews with 5, 4, 3, 2, 1 stars	"Thank you for your review" confirmation
TC_PROD_42 to 45	Direct URL loading, pricing visibility, stock availability, and meta title	Attributes verified

5.5 Shopping Cart Management Suite (50 Tests)

Class: `com.opencart.tests.ShoppingCartTest`

Test Method ID	Scenario Description	Expected Behavior
TC_CART_01 to 10	Add 10 products via @DataProvider (HP LP3065, in-stock, MacBook, iPhone, iPod, etc.)	Cart badge updates to show items
TC_CART_11 to 18	8 invalid/expired coupon codes via @DataProvider	Warning: "Coupon is either invalid, expired or reached its usage limit!"
TC_CART_19 to 26	8 invalid gift certificate vouchers via @DataProvider	Warning: "Gift Certificate is either invalid or the balance has been used up!"
TC_CART_27 to 30	Cart item table display and quantity updates to 2, 3, 5	"Success: You have modified your shopping cart!"
TC_CART_31 to 32	Item removal and empty cart display verification	"Your shopping cart is empty!" message
TC_CART_33 to 36	Shipping & tax estimation for US, UK, Canada, Australia	Quotes retrieved successfully
TC_CART_37 to 40	Totals table, proceed to checkout button, dropdown badge, and breadcrumbs	Components verified displayed
TC_CART_41 to 44	Coupon & voucher accordion expansion, empty continue redirect, and title	Accordions expand; redirects to Home
TC_CART_45 to 50	Single/multi-digit updates, promo input retention, direct URL, success banner	Modifications and alerts verified

5.6 Wishlist & Product Comparison Suite (42 Tests)

Class: `com.opencart.tests.WishlistAndComparisonTest`

Test Method ID	Scenario Description	Expected Behavior
TC_WSH_01 to 08	Add 8 products to wishlist as guest via <code>@DataProvider</code> (IDs 47, 43, 40, 30, 42, 33, 46, 28)	Alert prompts: "You must login or create an account to save..."
TC_CMP_09 to 16	Add 8 products to comparison matrix via <code>@DataProvider</code>	Success alert: "Success: You have added [Product] to your product comparison!"
TC_WSH_17 to 22	Wishlist counter, guest wishlist link redirecting to login, and notification links	Redirects to <code>route=account/login</code>
TC_CMP_23 to 33	Comparison table display: product image, name, price, model, brand, availability, rating, summary, dimensions	Comparison specifications verified displayed
TC_CMP_34 to 36	Remove product from comparison, empty comparison notice, continue button	"You have not chosen any products to compare."
TC_CMP_37 to 42	Side-by-side comparison of 2 products, breadcrumb, counter, title, clear all	Multi-column comparison matrix verified

5.7 End-to-End Guest Checkout Suite (40 Tests)

Class: `com.opencart.tests.CheckoutTest`

Test Method ID	Scenario Description	Expected Behavior
TC_CHK_01 to 06	Guest checkout complete order placement via @DataProvider (US CA, US NY, UK London, Canada ON, Australia NSW, Germany Berlin)	Full 6-step completion; returns "Order Placed"
TC_CHK_07 to 16	10 mandatory billing field validation tests via @DataProvider (missing country, zone, address)	Returns "Address Error" / validation alerts
TC_CHK_17 to 20	Step 1 guest checkout radio, Step 2 billing address form fields visibility	Accordion panels and inputs displayed
TC_CHK_21 to 23	Country dropdown AJAX population of State/Region (US states, UK counties)	Dependent zone dropdown options populated
TC_CHK_24 to 27	Step 4 flat rate shipping selection, Step 5 payment method & Terms & Conditions checkbox	Accordion transitions completed
TC_CHK_28 to 35	Order confirmation table, subtotal, shipping rate, final order placement, success heading	"Your order has been placed!" confirmed
TC_CHK_36 to 40	Empty cart checkout redirect, empty billing errors, UK and Canada full checkouts	Full order placement verified

5.8 Localization, Customer Service & Static Pages Suite (50 Tests)

Class: `com.opencart.tests.CustomerServiceAndStaticPagesTest`

Test Method ID	Scenario Description	Expected Behavior
TC_LOC_01 to 06	Currency switcher verification via @DataProvider (EUR, GBP, USD)	Active currency symbol updates to €, £, \$
TC_SVC_07 to 14	Contact Us validation rules via @DataProvider (empty name, short name, invalid email, empty/short message, valid enquiries)	Specific validation errors or enquiry sent success
TC_SVC_15 to 22	Product Returns validation rules via @DataProvider (empty name, email, telephone, order ID, product name, valid return)	Specific RMA errors or return confirmation
TC_LOC_23 to 25	Currency dropdown toggle, cart symbol synchronization, GBP persistence	Symbols update across storefront
TC_SVC_26 to 31	Contact Us store details, opening hours, Returns page reasons radio buttons	Customer service panels displayed
TC_INF_32 to 39	Information pages: About Us (4), Delivery (6), Privacy (3), Terms (5) and breadcrumbs	Content headings and breadcrumbs verified
TC_INF_40 to 42	Site Map page loading, category hierarchical tree, and breadcrumbs	Site map container displayed
TC_INF_43 to 46	Footer links: Brands/Manufacturers, Gift Certificates, Affiliates, Specials	Target pages loaded with valid titles
TC_INF_47 to 50	Footer copyright text, Information links, Service links, Account links	Footer structure verified

5.9 Admin Panel Access Suite (1 Test)

Class: `com.opencart.tests.AdminPanelTest`

Test Method ID	Scenario Description	Expected Behavior
TC_ADM_01	Admin panel login attempt and public demo logout tolerance	Gracefully handles login or maintenance lock without timeouts

6. Execution Commands & Reporting Guide

6.1 Command Line Execution (Maven)

All commands should be executed from the project root: `c:\Selenium\OpenCartAutomationTestNGSuite`

Execute Entire Suite (Headless Mode)

```
mvn clean test
```

Runs all 392 test cases in headless Chrome mode utilizing `src/test/resources/testng.xml`.

Execute in Visible Browser Mode (Headed UI)

```
mvn clean test -Dheadless=false
```

Launches visible Chrome windows for interactive debugging and visual inspection.

Cross-Browser Execution (Firefox / Edge)

```
mvn clean test -Dbrowser=firefox  
mvn clean test -Dbrowser=edge
```

Switches browser driver instances seamlessly through `DriverFactory`.

Execute a Specific Test Class or Single Test

```
# Run only Checkout test suite (40 tests)  
mvn clean test -Dtest=CheckoutTest  
  
# Run only Registration and Authentication suite (65 tests)  
mvn clean test -Dtest=RegistrationAndAuthTest  
  
# Run a single specific test method  
mvn clean test -Dtest=ProductSearchAndCatalogTest#TC_SRCH_26_searchPartialKeywordMac
```

6.2 Interactive HTML Test Reporting

Upon test completion, open the generated ExtentReports interactive dashboard in any browser:

```
test-output/ExtentReport/Index.html
```

Key report features include:

- **Visual Dashboard:** Graphical charts displaying Total Tests, Pass/Fail ratios, and execution duration.
- **Embedded Failure Screenshots:** If any test fails, an exact base64 screenshot of the browser state is automatically captured and embedded directly in the report log.
- **System & Environment Telemetry:** Records OS, Java runtime, framework version, and environment metadata.
- **Categorized Filtering:** Tests are tagged by test class for quick filtering and analysis.

Standard Surefire Reports:

Standard TestNG XML and HTML test reports are simultaneously generated in: `target/surefire-reports/index.html` and `target/surefire-reports/emailable-report.html`.